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Fomitopsis officinalis: a Species of Arboreal Mushroom with Promising Biological and Medicinal Properties

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Medicinal mushrooms of the order Polyporales have a long history of use, which is evidenced by the finding of dissected fruiting bodies with Ötzi, who lived over 5000 years ago.

Because of its valuable biological properties and its use in 18th and 19th–century pharmacy, *Fomitopsis officinalis* used to be mass–collected. Moreover, the large demand for larch wood and non–wood materials (resin) caused an excessive exploitation of larch forests, which directly contributed to the disappearance of *F. officinalis* from its natural environment.

The qualities of medicinal preparations obtained from the *F. officinalis* fruiting bodies are determined by the unique composition of its bioactive compounds, such as: triterpenoids, polysaccharides, organic acids, coumarins and phenolic compounds. It has been proved that both

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crude extracts and the compounds isolated from *F. officinalis* have a wide spectrum of therapeutic effects, including: anti–inflammatory, cytotoxic, and antimicrobial effects.

Keywords: agarikon, *Fomitopsis officinalis*, medicinal mushroom, triterpenoids.

1. Introduction

Fomitopsis officinalis (Vill.) Bondartsev & Singer (syn. *Laricifomes officinalis* (Vill.) Kotl. & Pouzar, *Fomes officinalis* (Vill.) Bres, *Polyporus officinalis* (Vill.) Fr. etc.) is a species of arboreal mushroom belonging to the family *Fomitopsidaceae*, Polyporales order. [1] Due to the rarity of its occurrence, it was first included on the list of 33 endangered fungal species in Europe and consequently listed as globally endangered on: "The IUCN Red List of Threatened Species" in 2019. [2] It is a parasite of coniferous trees of genus *Larix*, *Abies*, *Pinus*, *Picea*, *Cydrus*, *Tsuga* and *Pseudotsuga*. It causes the intensive brown rot of wood. [3] Because of its valuable biological properties and its use in 18th and 19th–century folk medicine, *F. officinalis* used to be mass–collected. Moreover, a large demand for larch wood and non–wood materials (resin, tan) caused the excessive exploitation of larch forests, which directly contributed to the disappearance of *F. officinalis* in the natural environment. [3] Researchers consider *F. officinalis* as one of the species of medicinal mushrooms most commonly used in traditional European medicine. Other examples include: *Fomes fomentarius*, *Fomitopsis pinicola*, *Laetiporus sulphureus* and *Piproporus betulinus*. [4]

F. officinalis used to be regarded as a panacea effective in treatment of numerous ailments, such as: excessive night sweating (flex. in tuberculosis patients), dizziness, rheumatism, respiratory, digestive and excretory diseases, and even cancer. [3,5] Hemorrhoids and

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dysmenorrhea were also treated with *F. officinalis* and the mushroom was believed to be a helminthagogue; large doses were to induce vomiting, while lower ones acted as diuretics. Indeed, both crude extracts and compounds isolated from *F. officinalis* were proven to have a wide spectrum of therapeutic effects, including: anti–inflammatory, [6] cytotoxic effects and antimicrobial effects. [4] It has been suggested that some of these properties were linked to the reactivity of triterpenoid compounds present in *F. officinalis*, although no plausible assessment, neither qualitative nor quantitative, of such an effect has been carried out. [7]

F. officinalis popularity in folk medicine lasted until the second half of the 19th century and the beginning of the 20th century. Soon after that it was progressively replaced by agaricin isolated from the mushroom itself. [4,8]

F. officinalis can be distinguished from other mushroom species due to its chalky texture, distinctive smell and taste, which makes it easily identifiable. [9]

The qualities of the crude material are a consequence of the composition of its bioactive compounds, such as: triterpenoids, polysaccharides, organic acids, coumarins and phenolic compounds. [10] Currently, due to its significant therapeutic potential, *F. officinalis* has been in the center of interest of numerous research teams.

The aim of this scientific publication was to present the current knowledge about *F. officinalis*, its biological and medicinal properties and the relevance of conducting clinical research, that might result in the discovery of new formulas, medicines or cosmetic preparations.

2. The Morphological Characteristics of *Fomitopsis officinalis* Fruiting Bodies

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Perennial (sometimes reaching as many as 75–80 old) *F. officinalis* fruiting bodies grow up to 1 m in high and may weight as much as 10 kg, accreting laterally to a tree trunk. They usually take a hoof–like shape, over time they become cylindrical, and the fruiting bodies intergrowth to finally form irregular clusters. The habitat area of *F. officinalis* extends from the roots up to the height of 20 meters on the tree trunk, sometimes reaching the branches, usually after several decades from the moment of infection. [3] The top surface is cream–colored, and becomes grey with age, it's cracked and widely furrowed, while the growth areas remain poorly marked. Algae often cover the mushroom with age. Its edges are rounded, and colored from fair through creamy to brownish. The tubes reach the length of 5–10 mm. The pores are usually angular or round, and with age they turn yellow and brown. [11]

The hyphae of *F. officinalis* mycelium are hyaline, which gives the mass a distinctive, whitish color. The flesh is usually white, caseous, soft, dry and brittle. Fully–grown specimens may contain splinters of wood and bark that were overgrown by the mushroom. *F. officinalis* is an inedible species, with floury, unpleasant smell. It is strongly bitter in taste. [1]

3. Habitat Characteristics

The *F. officinalis* natural range covers mainly the northern hemisphere. [3] Its habitats may be found in Africa (Morocco), Asia (China, India, Japan, Korea, Mongolia, Russia, Turkey), North America (USA, Canada) and Europe (Austria, Belarus, France, Greece, Spain, The Netherlands, Lithuania, Germany, Poland, Russia, Romania, Slovakia, Slovenia, Switzerland, Ukraine and Italy). [12] Currently, the species is most commonly found in Siberia, from the foothills of the Ural Mountains to the shores of the Pacific Ocean. [13] In Europe, *F. officinalis* resides on old

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larches, found mostly in natural forests, less often in the artificially–renewed forest stands. Currently, Polish habitats of *F. officinalis* consist of areas subject to legal protection, including reserves such as 'Modrzewina', 'Modrzewie', 'Świnia Góra', 'Ciechosławice', as well as the area of 'Chełmowa Góra' within the borders of the Świętokrzyski National Park. All stands are located within the Mazowieckie and Świętokrzyskie voivodships. Most of them cover the area of the Małopolska Upland. Another region of its occurrence is located in Wysoczyzna Rawska. [3,14]

4. *Fomitopsis officinalis* – History of Use

Due to their nutritional and medicinal properties, mushrooms have been used in traditional medicine of numerous cultures for ages, while the results of contemporary research attest to their remedial qualities. [15] The long history of use of medicinal mushrooms of the order Polyporales may be evidenced by the finding of dissected fruiting bodies of *Piptoporus betulinus* (Bull.) P. Karst. and *Fomes fomentarius* (L.) Fr., with Ötzi – 'the Iceman' – a natural mummy of a man who lived over 5000 years ago. [16]

In ancient Greece and Rome the mushroom was widely–known under the name of 'agarikon' or 'agaricum', mostly as an anti–tuberculosis agent. [12] First evidence of the use of 'agarikon' dates back as far as the 1st century, when Dioscorides – a Greek physician and philosopher – referred to it as a panacea for 'consumption'. [4] Another ancient Arabic physician – Ahwazi – defined cancer as a disease resulting from the accumulation of black bile and recommended that patients be given the raw mushroom material as a laxative (*purgativum*) in order to cleanse the body of bile. [17] The cleansing properties were reflected in the mushroom's name – *Boletus purgans* J.F. Gmel. 1792. [6]

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Agarikon (aka *gharikon*) obtained from *F. officinalis* is a material known and used outside the areas of its natural occurrence. Reports of its use in folk medicine may be found, among others, in the literature of Arabic countries, [17,18] Mongolia or Nigeria. [19,20] The species can also be found under the name of 'ghariqoun' in Iranian traditional medicine. [21]

In ancient times, the species was used in the treatment of pain, inflammation, jaundice, fever, insect bites, muscular diseases, bladder problems, sciatica, rheumatism and hemorrhoids. In addition, several authors emphasize its antiparasitic, anticoagulant and diuretic effects, as well as the ability to potentiate menstrual bleeding and increase immunity. [18,22,23] The species was also believed to be one of the most important anti–inflammatory agents. [6]

Fruiting bodies of *F. officinalis* were used in veterinary medicine. Based on the results of ethnobotanical research, it was found that 'agarikon' was a medicine used in Italy to prevent digestive disorders in cattle grazing the Western Alps. [6]

The *F. officinalis* fruiting bodies used to be called 'tree biscuits' and 'ghost bread'. Figurines made of the mushroom fruiting bodies were indispensable elements of rituals performed by the shamans of some Native tribes of North America. They were believed to have supernatural powers facilitating contact with spirits. Upon a shaman's death, the mushroom figurine was placed on his grave, to keep those who might approach at bay, as it was believed that the shaman's ghost resided inside. Many of such figurines – 'grave sentinels' – may be found in museums. [24]

Mushrooms used to play a particularly important role in the medicine of Russia and Slavic countries. The most important medicinal species of mushrooms used in these areas include *Inonotus obliquus* (Fr.) Pilát (Chaga), *Fomitopsis officinalis* (Vill.) Bondartsev & Singer, and *Fomes fomentarius* (L.) Fr. These species had been used to treat gastrointestinal problems,

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cancer, asthma or night sweats. [25] In the Middle Ages, the inhabitants of today's Ukraine and southern Russia would use the raw material of *F. officinalis* to medicate wounds sustained during battle [18] *The mushroom* was described as one of medical importance in the Cracovian Herbarium by Syrenius, published in 1613. [26] It was also one of the most important ingredients of laxatives and remedies popular in old Europe, included in e.g. *Theriac* or *Elixir ad longam Vitam*. [27] 'Agarikon' is also one of the ingredients of 'Swedish Bitters', the so–called Swedish remedies that have been popular across Europe since 1730. During this time the composition of the product had changed, but in its original form it contained seven ingredients: aloe juice, mustard seeds, turmeric, myrtle, medicinal rhubarb, saffron and 'agarikon'. [4,28] Preparations based on this species were also used in homeopathy, as the active ingredient of anti–migraine drugs. [29]

Although the mushroom was known and used in medieval Europe as early as 16th century, the widespread use of *F. officinalis* in European medicine took place over the course of the 19th century and up until the beginning of the 20th century. For instance, at this time the raw material was used to suppress night sweating in tuberculosis patients. [4,28] The crude material was available in pharmacies under the name of *agaryk*, *agarikon* or *Agaricus albus*. In Poland, the names Larch polypore or Larch sponge were also used. Agarikon was obtained from fruiting bodies that had been stripped of their hard outer layer. The crude material was sundried and sun–bleached, and then crushed into smaller fragments. To facilitate its further fragmentation, it was then submerged in tragacanth and dried, which would make the crude material more brittle and easier to pulverize. [27] This was used to make extracts, potions, pills, while the raw material was used to obtain resins: a red one, called *Agaricoresina* and a white one (aka white wax). In '*Pharmacognosia'* published in 1957 a component isolated from *F. officinalis* – agaricin (a

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tricarboxylic fatty acid) was described. The reported effects of agaricin were: the ability to decrease sweat secretion and stomach glands, and – in higher doses – stimulation of peristalsis in the gastrointestinal tract. By using various extraction methods of the fruiting bodies, extracts rich in agaricin were obtained, used in forms for both oral and subcutaneous administration. Due to its intensively bitter taste, the crude material was also used as an ingredient of the so–called bitter tincture, also as the quinine substitute and to falsify the quinin tincture *Tinctura chinae*.

Since up until the 20th century, *F. officinalis* was mass–harvested for commercial purposes, the popularity of this medicinal mushroom contributed to the depletion of its resources, and along with scarcity of *Larix* forest stands, increased the threat of extinction of the species. [3] The widely purported medicinal properties of *F. officinalis* along with its decline in natural environment strongly suggest, that the mushroom ought to be put under the scrutiny of contemporary clinical research, as this might result in the discovery of new formulas, medicines or cosmetic preparations. [18,22]

5. Biologically Significant Compounds of *Fomitopsis officinalis* Species

Although *F. officinalis* has been used for centuries, its fruiting bodies are still a source of new biologically active compounds, which might find use in modern medicine. The authors are well–aware that it is necessary to develop an effective method of both quantitative and qualitative analysis of the reactive compounds in the crude material, especially triterpenes. Nonetheless, scientific development on the chemical characteristics of metabolites and the *in vitro* assessment of their biological properties has been only recently developed. Moreover, only a few isolated

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compounds derived from *F. officinalis* have been studied individually, i.e. the biological activity of individual compounds has not yet been sufficiently determined. [30]

The main groups of active substances produced by mushrooms are, among others: polysaccharides, terpenoids, phenols and lectins. Over 126 therapeutic effects caused by these compounds have been demonstrated, which include immunomodulatory, antimicrobial, antiviral, antioxidant, hypercholesterolemic etc. [31,32]

F. officinalis fruiting bodies are characterised by a wide spectrum of properties, i.e. they have a warming effect, hence they eliminate phlegm and relieve asthma symptoms. [33] Polysaccharides isolated from medicinal mushrooms and their derivatives have become equally important in increasing the body's immunity which can be helpful in cancer treatment. [34] Additionally, polysaccharides obtained from *F. officinalis* are becoming more and more popular. [35,36] Polysaccharides and their complexes are perceived as valuable active substances obtained from mushrooms. [37] Due to their anti–cancer [38,39] and thei anti–inflammatory effects [18,40–42] they are used in medicine application. The researchers have characterized the structure of mannofucogalactan, the main ingredient of the *F. officinalis* fruiting bodies water extract. [43] Other important ingredients present in the crude material include, among others: 3–keto–dehydrosulfurenic acid, dehydrosulfurenic acid, dehydroeburicoic acid, fomitopsin C (Fig. 1) and the laricinolic acid.

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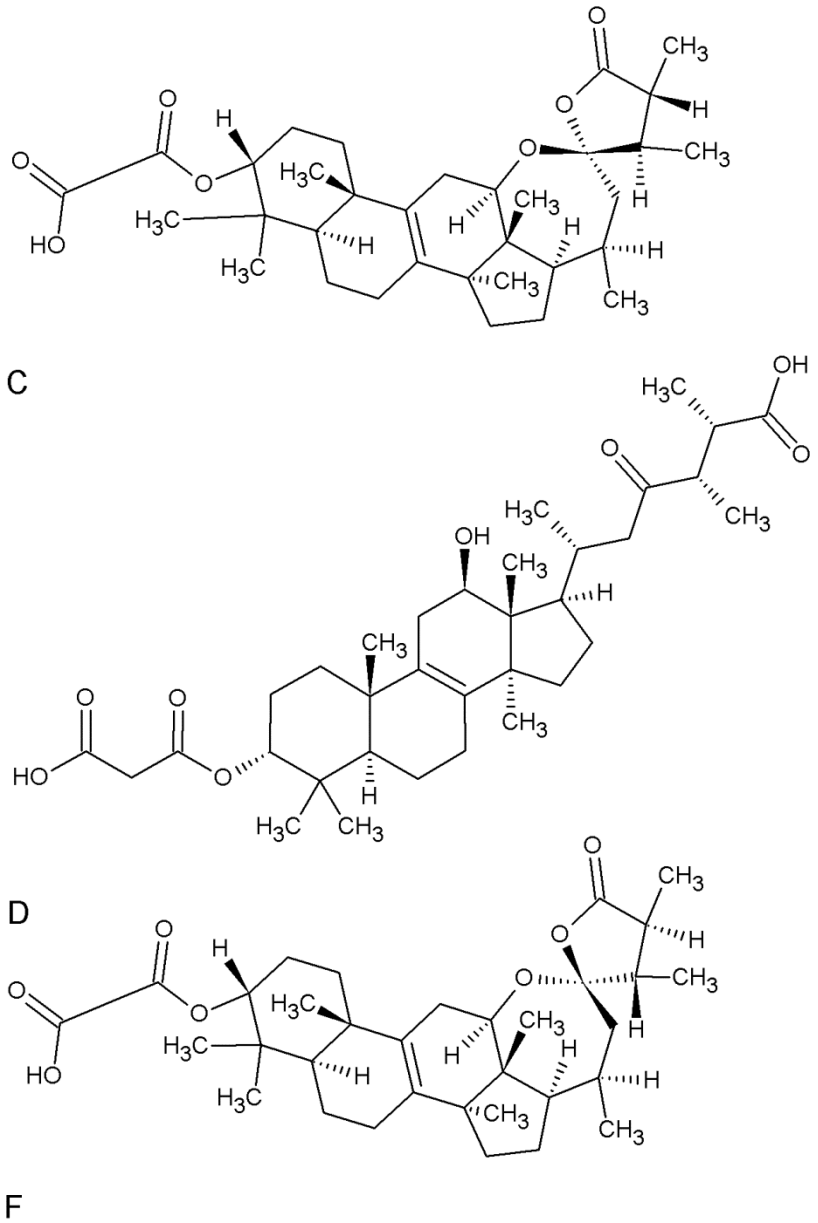

Figure 1. Chemical structures of fomitopsin C, D, F

Researchers also report new compounds, such as: triterpenoid lactone – fomefficin or a sesquiterpenoid – fomefficinic acid. [44] According to Wu and co–workers, terpenes and steroid compounds have been identified as the main active substances of the species, and are responsible for the observed crude material's effects. [45] The researchers have isolated two new terpenoid

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acids of the lanostane type from the methanol–based *F. officinalis* extract, i.e. 12β,15α–dihydroxy–24–methyl–3,23–dioxo–lanosta–7,9(11)–dien–26–oic acid, and 3α, 12β–dihydroxy–24–methyl–7,23–dioxo–lanosta–8–en–26–oic acid. [46] Furthermore, several new triterpene compounds have been isolated from the fruiting bodies of this species, including: fefresin F and G acids, fomefficinol A and B (Fig. 2.), fomlactone A and B, ergosterol, laricinolic acid, ergota–7,22–dien–3–beta–ol together with seven known officinalic acids.

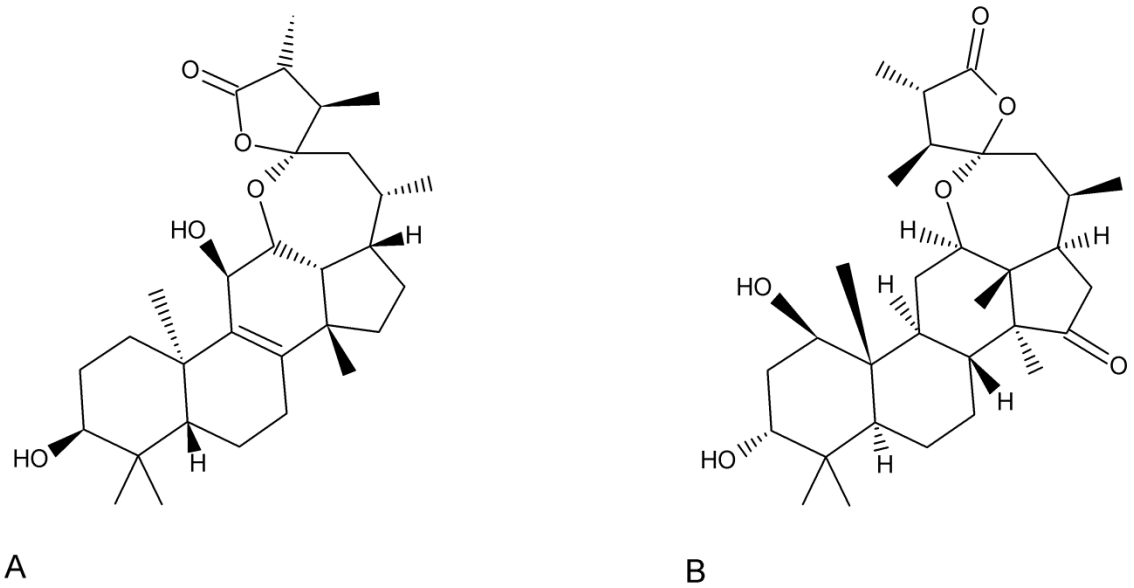

Figure 2. Chemical structures of fomefficinol A and B

The structures of newly–isolated compounds were determined by means of chemical methods and spectroscopic analyses. [47] Five new triterpenes of the lanostane type called the fomefficinic acid A–E (Figure 3) were isolated from the powdered *F. officinalis* mycelium. [48–50]

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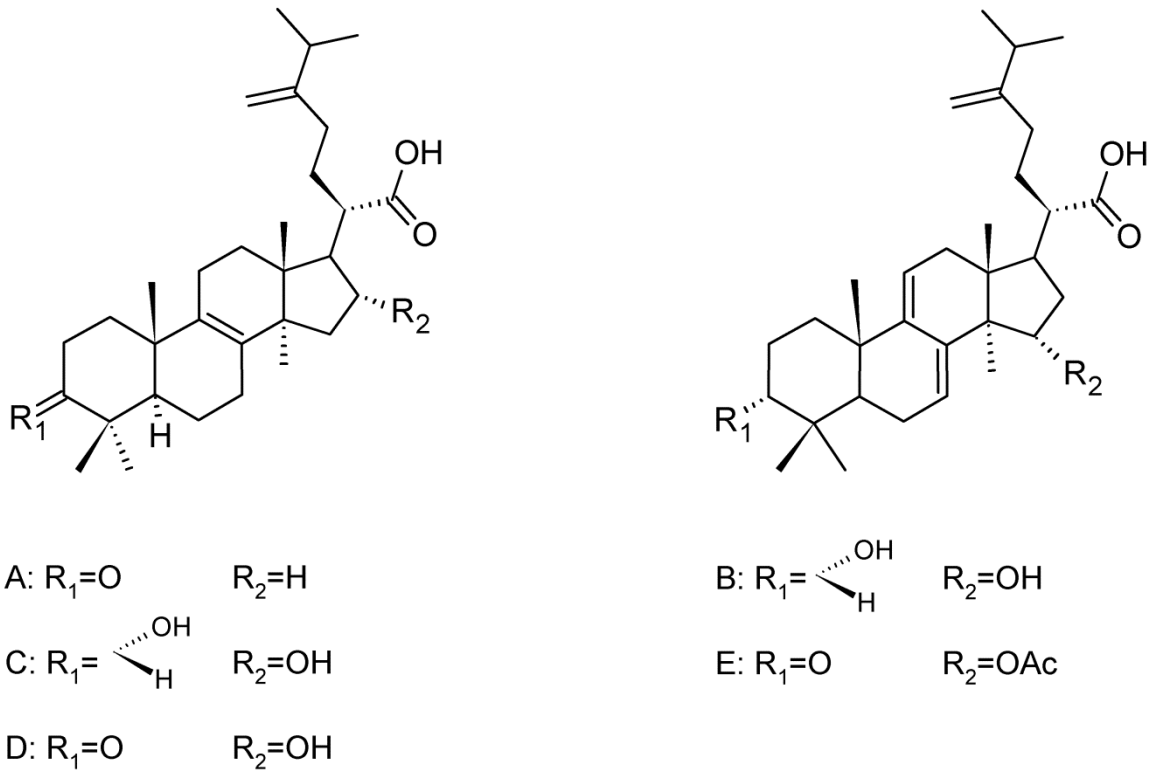

Figure 3. Chemical structures of fomefficinic acids A–E

Table 1 briefly lists compounds described in the literature and their biological activity.

Table 1. Secondary metabolites of *Fomitopsis officinalis* and their biological properties

Chemical compounds	**Biological properties**	**References**
Polisaccharides	immunomodulatory, anticancer, anti-inflammatory	[34], [39], [38], [40], [51], [41]
3-Keto-dehydrosulfurenic acid	anticancer	[52]
Dehydrosulfurenic acid	anticancer, neuroprotective	[52], [53]
Dehydroeburicoic acid	anticancer	[52]
Eburicoic acid	anti-inflammatory	[54]
Fomitopsin C	anticancer	[52]

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Fomitopsin D, E, F	antiviral	[30]
Triterpenoid lactone – fomefficin	anticancer	[44]
Sesquiterpenoid – fomefficinic acid	anticancer	[44]
eight lanostane-type terpenoids	antiviral	[19], [30]
6-Chloro-4-phenyl-coumarin	antimicrobial	[55]
2H-6-chloro-2-oxo-4-phenyl-1-Benzopyran-3-carboxylic acid ester	antimicrobial	[55]
Triterpenoids of lanostane type	antiviral	[18]
Officimalonic acids A – H	anti-inflammatory anticancer cytotoxic	[56]
24-Dien-21-oic acid, 3-acetyloxylanostane-8	anticancer	[52]
3-Keto-dehydrosulfurenic acid	anticancer	[52]
Dehydroeburicoic acid	anticancer	[52]
Succinic acid	anticancer	[52]
Fomitopsin C	anticancer	[52]
Pinicolic acid	anticancer	[52]
Trametenolic B acid	anticancer	[52]
Flavonoids	antioxidative	[57]

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5. 1.The Anti–Microbial Activity of *Fomitopsis officinalis*

Mushrooms are a valuable source of active substances, with antibacterial activity. Some of these metabolites have evolved as a response against a wide spectrum of microbial antagonists. This makes them particularly interesting as possible future pharmacological agents. [32,58,59] From a scientific point of view, species recognized as medicinal in traditional medicine, are more effective when administered separately, than as a component of multi–ingredient preparations. [30] A number of studies have been carried out to test the biological activity of extracts and compounds isolated from *F. officinalis*, which proved antimicrobial activity. In the study of raw ethanol extracts of four medicinal species of basidiomycetes and their effects on Gram(+) (*Bacillus thuringiensis, Staphylococcus aureus*) and Gram(–) (*Klebsiella pneumoniae, Enterobacter aerogenes*) bacteria it was proven that only *F. officinalis* was capable of inhibiting all those bacterial strains, and its MIC (minimum inhibitory concentration) was very high. It has been reported that *F. officinalis* extracts present a moderate activity against some bacterial strains. [19,44] Hwang and co–authors used *F. officinalis* mycelium to isolate two chlorinated coumarin derivatives. [55] The biological activity of coumarins can be characterised by a wide spectrum of pharmacological effects, such as: antiviral, antibacterial, antifungal, antiparasitic and analgesic. It is also believed that chlorination and bromination are the two factors responsible for increasing the antimicrobial activity against *Vibrio sp.*, [60–62], which can be explained by the immunotropic activity exhibited by its extracts. [63] The authors would also like to accentuate the inhibitory effect of coumarin compounds isolated from the *F. officinalis*, against the following bacteria strains: *Agrobacterium tumefaciens, Escherichia coli, Bacillus subtilis, B. cereus,*

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Salmonella typhimurium, *Staphyloccocus aureus*, as well as its weaker inhibition of *Candida albicans* and *Acinetobacter baumanii*, *Enterococcus faecalis*, *Pseudomonas aeruginosa*, *Mycobacterium smegmatis* as well as *Streptococcus pneumoniae*. [55]

The anti–tuberculosis effect of the raw material has also been described. [64] Due to the increasing resistance to antibiotic therapy, other medicinal substances are still being sought. Coumarins isolated from the *F. officinalis* ethanol extract exhibit activity against *Mycobacterium tuberculosis*. [56] Studies on anti–tuberculosis properties indicate a significant potential of coumarin compounds, particularly 6–chloro–4–phenyl–coumarin and 2H–6–chloro–2–oxo–4–phenyl–1–benzopyran–3–carboxylic acid ester, isolated from *F. officinalis*. [4] The research conducted so far proved the antibacterial activity of the species' mycelium against Gram(–) bacteria. On this basis it was hypothesized that the *F. officinalis* species could be used as a treatment for pseudotuberculosis, as well as against the bacteria of the genus *Pseudonomas*. [65] *In vitro* research conducted in the USA proved that almost 100 species of mushrooms exhibit strong antiviral properties, particularly against the chickenpox virus. [6] A study based on the use of *F. officinalis* in Iranian medicine confirmed its antiviral effect, particularly against smallpox, influenza subtype H5N1 and hepatitis C. [18] The species' activity results mostly from the content of triterpenoids of lanostane type. Another study also indicated the activity of fomitopsin D (Figure 1) against herpes simplex virus type 1 (HSV–1), and fomitopsin F (Figure 1) against *B. cereus*. [30]

Researchers studying *F. officinalis* fruiting bodies obtained in Altai, showed that the antiviral properties of the fruiting bodies extend to influenza virus types A/chicken/Kurgan/05/2005 (H5N1) and A/Aichi/2/68 (H3N2). A similar effect has been confirmed for several other mushroom species. [66] Based on the activity study of eight lanostane–type terpenoids isolated

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from *F. officinalis*, it may be concluded that those substances have an inhibitory effect on *Trypanosoma congolense* (nagana trypanosome). This trypanosome, being one of the most pathogenic parasites found in Mongolia, causes fatal diseases in animals, mostly farm livestock, which is the main source of income for the majority of the population. Therefore, this information seems extremely valuable when it comes to the medicinal use of the species, which, however, requires further detailed study. [20] In this regard it is worth mentioning, that direct antiviral activity of this species against cowpox (*Orthopoxvirus*) has been described. [30]

As a little as 1–2% of *F. officinalis* extract is able to inhibit virus–induced cell damage by 50%. The crude extract diluted in a ratio of 1:106 is still effective against type A and B viruses and the herpes virus. [30] Many experiments have proved the antimicrobial properties of *F. officinalis*, indicating its potential use as a raw material for the production of antiviral, antibacterial and antiparasitic drugs. This activity profile is shared by coumarins and triterpenes, therefore the effectiveness of *F. officinalis* against other microorganisms in comparison to other fungal species requires further study. Although not many fungal species exhibit higher antimicrobial activity, *F. officinalis* may prove competitive when used for specific purposes, for example against *Mycobacterium tuberculosis* or *Yersinia pseudotuberculosis*. [30]

5.2. Antioxidant Activity and Anti–Cancer Properties of *Fomitopsis officinalis*

The excess of oxidizing factors disturbs the balance between the level of oxidative stress and oxidant activity, contributing to the development of numerous diseases. These include, but are not limited to: cancer, atherosclerosis, cardiovascular disease, infections and allergies, as well as

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premature ageing of the body. Therefore, providing antioxidants with food is justified to support the body's natural defenses. [50]

Fruiting bodies of mushrooms are a rich source of phenolic compounds with proven antioxidant and cytotoxic effects. The effect of phenolic compounds isolated from *F. officinalis* anti-senile action has been demonstrated in *in vivo* studies. Three groups of mice were administered different doses: 100 mg/kg, 200 mg/kg and 400 mg/kg of flavonoids obtained from the raw material. It was found that each of the administered doses of *F. officinalis* flavonoids may, to a various extent, increase the index of antioxidative activity in the brain, spleen and thymus (glutathione peroxidase activity (GSH–Px) in brain tissues, catalase (CAT) and superoxide dismutase (SOD) in liver tissues). [57]

In recent years, it has been proved that some of the higher mushroom species inhibit the growth of cancer cell lines. So far, approximately two hundred species have been identified, exhibiting this activity, [66] with *F. officinalis* being one of them. Many new compounds isolated from the raw material are believed to have an effect on cancer cells. Studies involving triterpene lactone – fomefficin, and fomefficin sesquiterpene have shown their significant anti–cancer effect. [44] Similar properties have been observed in a group of triterpene compounds called the officimalonic acids A – H (eight triterpenes of 24–methyl–lanostane) isolated from the methanol extract of *F. officinalis*. Their anti–inflammatory and cytotoxic activity in *in vitro* conditions was confirmed in contact with human cancer cells H460, HepG2 and BGC–823. [46] In a study comparing the anti–cancer effect of several species of mushrooms, alcohol and water extracts of the following mushrooms were evaluated: *Ganoderma sinense, F. officinalis, F. pinicola, Polyporus melanopus* and *Taiwanofungus camphorates*. The anticancer effect of all ethanol extracts was proved in *in vitro* conditions, thus noting the lower activity of aqueous extracts of

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the tested raw material. The highest activity was recorded for the ethanol extract of *F. pinicola*. [45] Moreover, the results of seven remaining triterpene compounds of lanostane type, obtained from *F. pinicola* and *F. officinalis*, such as: 24–dien–21–oic acid, 3–acetyloxylanostane–8, 3–keto–dehydrosulfurenic acid, dehydroeburicoic acid, succinic acid, fomitopsin C, pinicolic acid and trametenolic B acid testify to the anticancer properties of those compounds in *in vitro* conditions. [52]

It has been found that both the antioxidant and cytotoxic activities against lines of human cancer cells (incl. small–cell lung carcinoma, brain astrocytoma, lung adenoma and colorectal cancer) are higher in case of fruiting body extracts of several species, than in single species extracts. [28]

5.3. The Potential Influence of Biologically Active Compounds of *Fomitopsis officinalis* on the Central Nervous System

Among the various possible applications of mushrooms and their bioactive compounds in medicine, the improvement of central nervous system's functions has only recently gained attention of the scientific community. It is however already apparent, that several species of mushrooms stimulate the production of neuronal growth factors (ex. the Nerve. Growth Factor, NGF; the brain–derived neurotrophic factor, BDNF) (*ibidem*) and show therapeutic potential in neurodegenerative disorders, [67] as well as mood disorders. [68,69] Moreover, it has recently been shown that the memory–enhancing effects of 2 months of dietary supplementation with *Hericum erinaceus* are present even in healthy animals, [70] which extends the possible neurological benefits of mushroom supplementation to the entire population.

To date, there are only a few studies directly linking the bioactive compounds of *F. officinalis*

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with brain functions. By administering flavonoids daily for 6 weeks in doses ranging from 100 to 400 mg/kg, Sha et al. 2016 has shown that *F. officinalis* can counteract oxidative stress in the aging brain of mice. [57] Due to its high energy and oxygen demands (~20% of the oxygen available in the human body is consumed by the brain), nervous tissue is particularly vulnerable to overproduction of reactive oxygen species, which can lead to neurodegenerative disorders such as Alzheimer's disease, multiple sclerosis, Huntington's disease or Parkinson's disease. [71] However, the implications of antioxidative activity of *F. officinalis* in the brain reach beyond neurodegeneration, since oxidative insults are also a factor in the etiology of both epilepsy [72] and clinical depression. [73] Moreover, dietary flavonoids are known to exert their action in the brain by directly regulating a group of intracellular kinases, that converge on a transcription factor (cAMP response element–binding protein, CREB) crucial for modulating the strength of connections between neurons (i.e. synaptic plasticity). [74] One of the ways in which CREB affects synaptic plasticity and neuronal survival is by regulating the level of the BDNF protein. Therefore, it is possible that the BDNF–mediated antidepressant [68] and pro–cognitive [70] effects observed after supplementation with other mushrooms, are driven by phenolic compounds, and therefore could be obtained with *F. officinalis*. Recent studies also point to the flavonoid–induced CREB–BDNF pathway activation as a mechanism responsible for the anti–epileptic activity of flavonoids. [75]

Among the bioactive compounds of *F. officinalis*, lanostane–type triterpenoids constitute the biggest group. [4] Recently, one of these compounds – found specifically in *F. officinalis* (dehydrosulfurenic acid) – has been patented as a potential pharmaceutical treatment for ischemic stroke. [53] At a dose of 50 mg/kg, dehydrosulfurenic acid administered 10 minutes before an ischemic insult, significantly reduced the level of motoric deficits and neuronal injury

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monitored in rats for up to 24 hours after the injury. Eburicoic acid, which is another triterpenoid found in *F. officinalis*, with anti–inflammatory properties, that could also be neuroprotective. When tested in macrophages cell lines, eburicoic acid inhibits proinflammatory cytokines (interleukin–1β, interleukin–6, tumor necrosis factor and nitric oxide), without any cytotoxic effects. [54] This profile of action is identical to the one recently observed in microglia (i.e. resident macrophages of the brain) exposed to a different, naturally occurring triterpenoid – platycodigenin, which promotes regeneration of neurons. [76] Moreover, since all of the aforementioned proinflammatory cytokines are upregulated in clinical depression, [73] eburicoid acid could further prove to have antidepressant effects. Triterpenoids are also known to activate peroxisome proliferator–activated receptors (PPARs) [76] – a quality shared by a growing list of mushrooms, [77–79] which could prove effective in treating neurodevelopmental disorders, such as autism spectrum disorder. [80]

Lastly, it has been suggested that the vast anti–viral effects of *F. officinalis* could prevent neuropathies associated with infections caused by the herpes viruses or hepatitis C. [81] Surely, further research will elucidate the full potential of *F. officinalis* in regulating central nervous system's functions and hopefully point to novel mechanisms, which could be used to improve the treatment of this complex organ.

6. *Fomitopsis officinalis* and its use in medicinal and dietary preparations

Pharmaceutical, food and cosmetic industries show a growing interest in obtaining mushrooms preparations *in vitro*. Many medicines commonly used all over the world (Koriolan, Krestin, Lentinan, Schizophyllan etc.) contain substances of fungal origin. Most of these

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preparations have a beneficial influence on health due to the synergistic effect of the active substances contained in them, while maintaining the safety of their use. [82] The nutritional, anti–inflammatory, regenerative and antioxidant properties of numerous mushrooms' raw materials mean there is a perspective for their wider use in the production of cosmetics. [42,51,83] Mushrooms obtained by means of biotechnological methods have also found their use as ingredients in balanced animal foods. [84] Due to the threat of extinction and conservation status, *F. officinalis* is rarely used in medicine.

Preparations containing powdered *F. officinalis* mycelium are, however, available on the Internet both in Europe and the USA. Some products appear under the name of the raw product commonly used in 19th and 20th centuries, which is not reflected in their composition, and therefore can only by considered as a 'marketing ploy'. Frequently, the name 'Agarikon' is actually a preparation whose main ingredient is the powdered mycelium of *Agaricus blazei*, *Ganoderma lucidum* or *Lentinus edodes*. [85] Very often, the raw material is but one of a few, several or even several dozen ingredients of the supplements. Recently, mushroom–based preparations have been offered in the form of vegetarian or vegan capsules.

Drops, capsules or concentrates that strengthen the body and improve liver or bile duct functions, based on the properties of *F. officinalis*, and originating from Ukraine, have been available in online stores. 'Agarikon' capsules and powdered *F. officinalis* mycelium are sold in the USA and online.

7. Conclusion

This work describes the *F. officinalis* fruiting bodies and their ability to synthesize metabolites with potential use in medicine, cosmetology and production of pharmaceutical,

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dietary and cosmetic preparations. Using the extracts of this species, and especially their biomass from *in vitro* cultures, might prove an important avenue in the continuing search for the cure for civilization diseases.

Disclosure statement

No potential conflict of interest was reported by the authors

Author Contribution Statement

B. M. designed and directed the project, B.M.; K. Z-S; J. P.; P.K. and A.F. and analyzed the data and wrote the paper; A.W. and E. N. collected the literature data; All authors discussed the results and contributed to the final manuscript.

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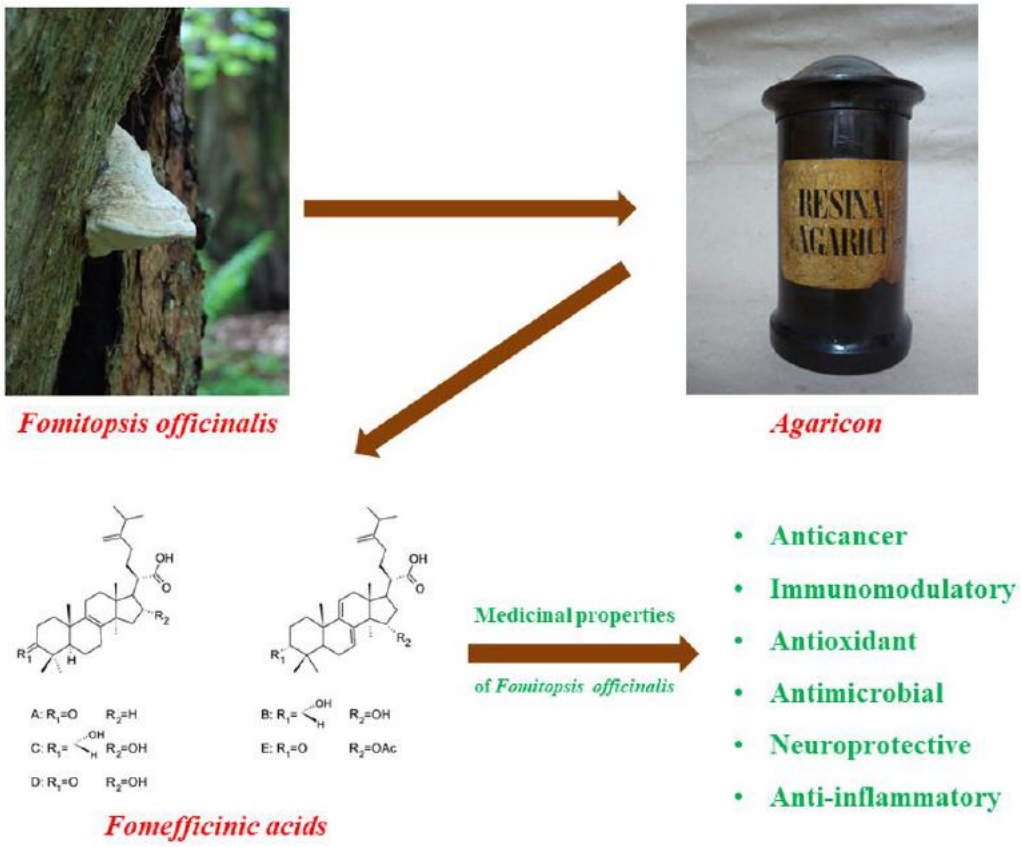

Bożena Muszyńska, professor of medical science at the Faculty of Pharmacy of the Jagiellonian University Medical College. The most important aim of her research is to prove the therapeutic and dietary value of mushrooms (Basidiomycota), while the aim of research in the field of mushroom biotechnology is to propose mycelial cultures as a rich, alternative source for bioactive components obtaining. Another subject of research was an effect of edible mushrooms fruiting bodies processing on the content of selected biologically active compounds and their supplementation by mycelia in active substances in order to obtain a material with better dietetic and medicinal properties.

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An important direction of her research in the field of mycochemistry is the assessment of biological activity (inter alia: anti-inflammatory, antioxidant, cytotoxic) of extracts obtained from the fruiting bodies and mycelium of the examined species. Another interests were also focused on the evaluation of mycelium from *in vitro* cultures in terms of bioremediation of the natural environment from xenobiotics by selected mushroom species. She authored more than 140 articles in peer-reviewed international journals, 4 books, and 14 chapters in textbooks and many invited conference.

Agata Fijałkowska attained her MA in Psychology, followed by a BSc in Cultural Anthropology and another MA in Neurobiology in 2019. Currently she is a PhD candidate at the Doctoral School of Medical and Health Sciences of the Jagiellonian University Medical College in Kraków. She investigates the impact of organic compounds found in medicinal mushrooms on the functioning of the central nervous system and conducts her research at the Department of Pharmaceutical Botany.

Katarzyna Sułkowska-Ziaja is an assistant professor at the Faculty of Pharmacy at the Jagiellonian University. She is the author of over 70 peer-reviewed international articles and several book chapters. Her research interests focus mainly on the analysis of bioactive compounds accumulated in the fruiting bodies of medicinal mushrooms and on biotechnology of higher fungi. Her research concerns, among others, optimization of mycelium cultures conditions and the possibility of obtaining active metabolites throughout the stimulation of their endogenous biosynthesis and the use of precursors and stimulators of metabolic pathways.

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Anna Włodarczyk has master's degree in Pharmacy and is currently a Ph.D. student at the Faculty of Pharmacy of the Jagiellonian University. She's the author of several publications, including a book chapter, numerous scientific articles and reports at international and national conferences. Moreover, she has been conducting trainings for pharmacists. Professionally, she works as a director of the pharmacy in international network Super-Pharm.

Piotr Kaczmarczyk (PhD in Pharmaceutical Botany 2006). Hospital pharmacist working in the field of drug logistic and parenteral nutrition. Currently work as a quality auditor and medical drug management coordinator in the Provincial Specialist Hospital in Częstochowa.

Vice President of the Silesian Pharmaceutical Chamber, lecturer at the Higher School of Medicine in Sosnowiec, and also publicist (over 200 popular science articles, 8 monographs). In the years 2006-2008, he performing duties as a president of the Polish Pharmaceutical Society in Kielce site. Owner of a training and research company "INTERMEN Piotr Kaczmarczyk". The author's professional interests include mainly issues related to the quality and safety of pharmacotherapy, clinical malnutrition, and pharmaceutical care. Moreover, the author actively participates in many research projects in the field of natural medicine included medical plants and mushrooms.

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Ewa Nogaj – PhD in medical sciences. Dean of the Faculty of Health Sciences at the Higher School of Medicine in Sosnowiec. A longtime employee of the Department of Toxicology of the School of Pharmacy with Division of Laboratory Medicine of Medical University of Silesia in Katowice. Her areas of scientific interest are: health sciences, medical sciences, toxicology, environmental protection, environmental monitoring, and in particular the use of various types of biological samples, e.g. pharyngeal tonsils, arteries; environmental samples (water, soil, air), samples of plant materials (fungi, medicinal plants) as biomarkers of human exposure to heavy metals. Author of 79 scientific publications, 40 popular science articles, 2 monographs and 39 congressional announcements.

Jacek Piętka is an employee of the Department of Forest Protection at the Institute of Forest Sciences, Warsaw University of Life Sciences – SGGW since 1998. He received a PhD in forest sciences in 2003. He is the author of 40 publications in peer-reviewed scientific journals. His research interests are mainly focused on forest pathology, mycology, nature protection, including active protection of endangered wood-decay fungi in forests. He is also interested in beetles

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associated with fungi. His research concerns both biotic and abiotic factors affecting the health of a forest ecosystem.

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